

## 應數二離散數學 2026 春, 期末考

學號: \_\_\_\_\_, 姓名: \_\_\_\_\_

本次考試共有 13 頁 (包含封面), 有 13 題。如有缺頁或漏題, 請立刻告知監考人員。

### 考試須知:

- 請在第一及最後一頁填上姓名學號。
- 不可翻閱課本或筆記。
- 計算題請寫出計算過程, 閱卷人員會視情況給予部份分數。  
沒有計算過程, 就算回答正確答案也不會得到滿分。  
答卷請清楚乾淨, 儘可能標記或是框出最終答案。

### 高師大校訓: 誠敬宏遠

誠, 一生動念都是誠實端正的。      敬, 就是對知識的認真尊重。  
宏, 開拓視界, 恢宏心胸。      遠, 任重致遠, 不畏艱難。

請尊重自己也尊重其他同學, 考試時請勿東張西望交頭接耳。

1. (10 points) Consider the following 6-by-6 board with forbidden positions (marked with x). Determine the number of ways to place six non-attacking rooks on the allowed positions. (不需化簡)

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| x | x |   |   |   |   |
|   | x |   |   |   |   |
|   |   | x | x |   |   |
|   |   |   | x |   |   |
|   |   |   |   | x | x |
|   |   |   |   |   |   |

Answer: \_\_\_\_\_

2. (10 points) Given a sequence with  $h_0 = 2$ ,  $h_1 = 3$ ,  $h_2 = 10$ , and the third difference is constant:

$\Delta^3 h_n = 18$ . Determine  $h_3$ , the expression of  $h_n$ , and a formula for  $\sum_{k=0}^n h_k$ . (不需化簡)

Answer:  $h_3 =$  \_\_\_\_\_;  $h_n =$  \_\_\_\_\_;  $\sum_{k=0}^n h_k =$  \_\_\_\_\_.

3. (10 points) We want to tile a  $2 \times n$  board using two types of tiles: (1) a  $1 \times 2$  domino (which can be placed vertically or horizontally), and (2) a  $2 \times 2$  square tile. Let  $h_n$  be the number of ways to tile the  $2 \times n$  board.

(a) Find the recurrence relation for  $h_n$  and state the initial conditions  $h_1$  and  $h_2$ .

Answer:  $h_n =$  \_\_\_\_\_,  $h_1 =$  \_\_\_\_\_,  $h_2 =$  \_\_\_\_\_.

(b) Solve the recurrence relation to find an explicit formula for  $h_n$ .

Answer:  $h_n =$  \_\_\_\_\_

4. (10 points) Solve the nonhomogeneous recurrence relation  $h_n = 4h_{n-1} - 4h_{n-2} + 3^n$  with initial values  $h_0 = 10$  and  $h_1 = 35$ .

(a) The form of the particular solution is  $h_n^{(p)} =$  \_\_\_\_\_.

(b) The general solution is  $h_n =$  \_\_\_\_\_.

5. (10 points) Determine the number of integral solutions of the equation

$$x_1 + 5x_2 + x_3 + x_4 = n, \text{ that satisfy } 0 \leq x_1 \leq 4, 0 \leq x_2, 4 \leq x_3, -1 \leq x_4$$

(a) The generating function  $G(x) =$  \_\_\_\_\_.

(b) The coefficient of  $x^n$  in  $G(x)$ , i.e.  $h_n =$  \_\_\_\_\_.

6. (10 points) Let  $h_n$  denote the number of  $n$ -digit numbers with all digits at least 4 (i.e. digits 4, 5, 6, 7, 8, 9), such that 5 occurs an even number of times, 7 and 8 each occur an odd number of times, 9 occurs at least once, and there is no restriction on the remaining digits (4 and 6). Determine the generating function  $g(x)$  for the sequence  $h_0, h_1, h_2, \dots$  and then find a simple formula for  $h_n$ .

Answer: (a) The generating function  $g(x) =$  \_\_\_\_\_,

(b) The coefficient of  $\frac{x^n}{n!}$  in  $g(x)$ , i.e.  $h_n =$  \_\_\_\_\_.

7. (8 points) The number of partitions of a set of  $n$  elements into  $k$  distinguishable(可區分的) boxes (some of which may be empty) is  $k^n$ . By counting in a different way, prove that

$$k^n = \binom{k}{1} 1! S(n, 1) + \binom{k}{2} 2! S(n, 2) + \dots + \binom{k}{n} n! S(n, n)$$

(If  $k > n$ , define  $S(n, k)$  to be 0.) Here  $S(n, k)$  are the Stirling number of the second kind.

8. (7 points) Let  $s(n, k)$  be the Stirling numbers of the first kind. Find  $s(5, 3) = \underline{\hspace{2cm}}$ .

9. (5 points) Find the Ferrers diagram of the given partition  $\lambda : 24 = 8 + 6 + 4 + 3 + 2 + 1$ , and then determine the conjugate partition  $\lambda^*$ .

Answer:  $\lambda^* =$  \_\_\_\_\_

10. (10 points) Let  $p_n^s$  equal the number of self-conjugate partitions of  $n$ . Find  $p_{16}^s$ . *Hint:* By Theorem 8.3.2,  $p_n^s = p_n^t$ , where  $p_n^t$  is the number of partitions of  $n$  into distinct odd parts.

Answer:  $p_{16}^s =$  \_\_\_\_\_

11. (10 points) Let  $f_n$  is the  $n^{th}$  Fibonacci number. Prove that  $f_n$  is divisible by 3 if and only if  $n$  is divisible by 4.
12. (5 points) Construct a linear homogeneous recurrence relation with constant coefficients for which  $h_n = n^2$  is a solution (state the recurrence and a set of initial conditions).

Answer: \_\_\_\_\_.

13. (10 points) Let  $m$  and  $n$  be nonnegative integers with  $n \geq m$ . There are  $m + n$  people in line to get into a theater for which admission is 50 cents. Of the  $m + n$  people,  $n$  have a 50-cent piece and  $m$  have a \$1 dollar bill. The box office opens with an empty cash register. Show that the number of ways the people can line up so that change is available when needed is

$$\frac{n-m+1}{n+1} \binom{m+n}{m}$$

學號: \_\_\_\_\_, 姓名: \_\_\_\_\_, 以下由閱卷人員填寫

[illegible]